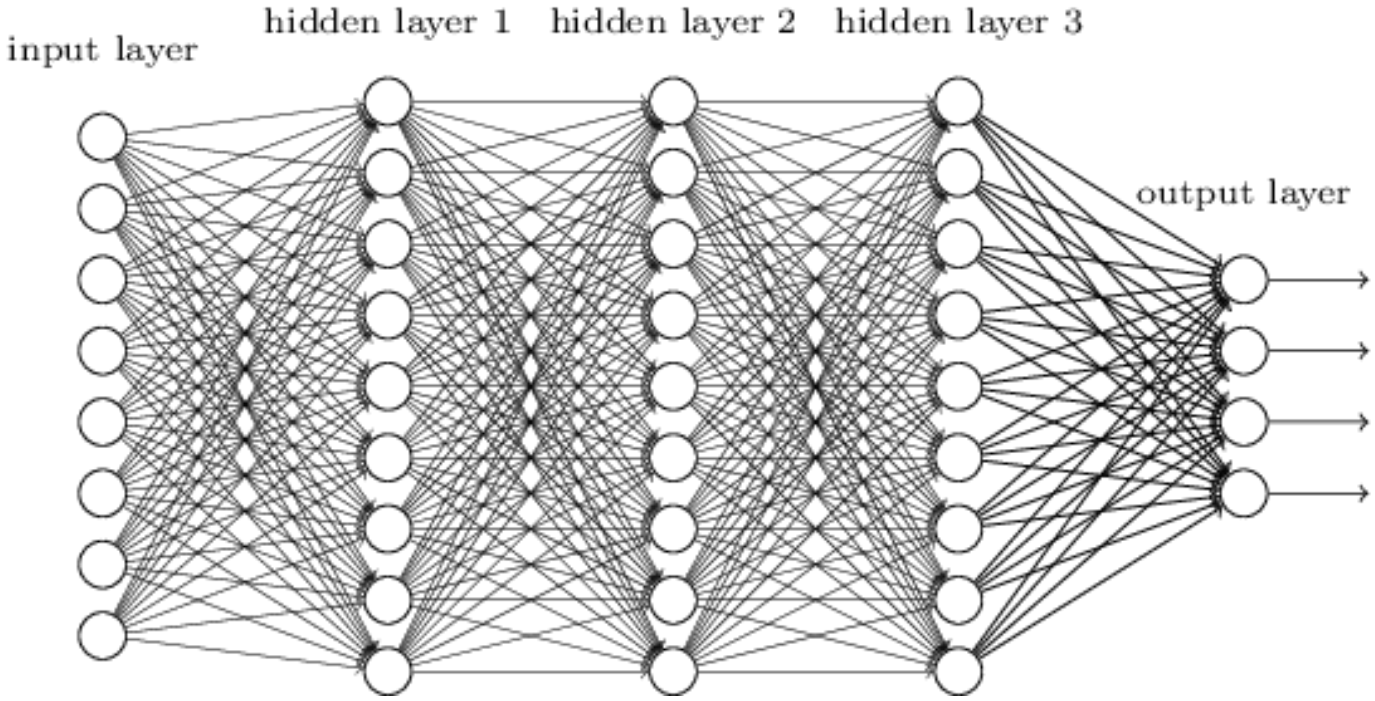
Smaller Network: CNN

- We know it is good to learn a small model.
- From this fully connected model, do we really need all the edges?
- Can some of these be shared?



Consider learning an image:

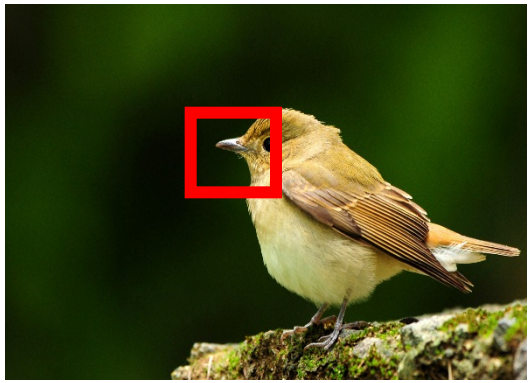
- Some patterns are much smaller than the whole image

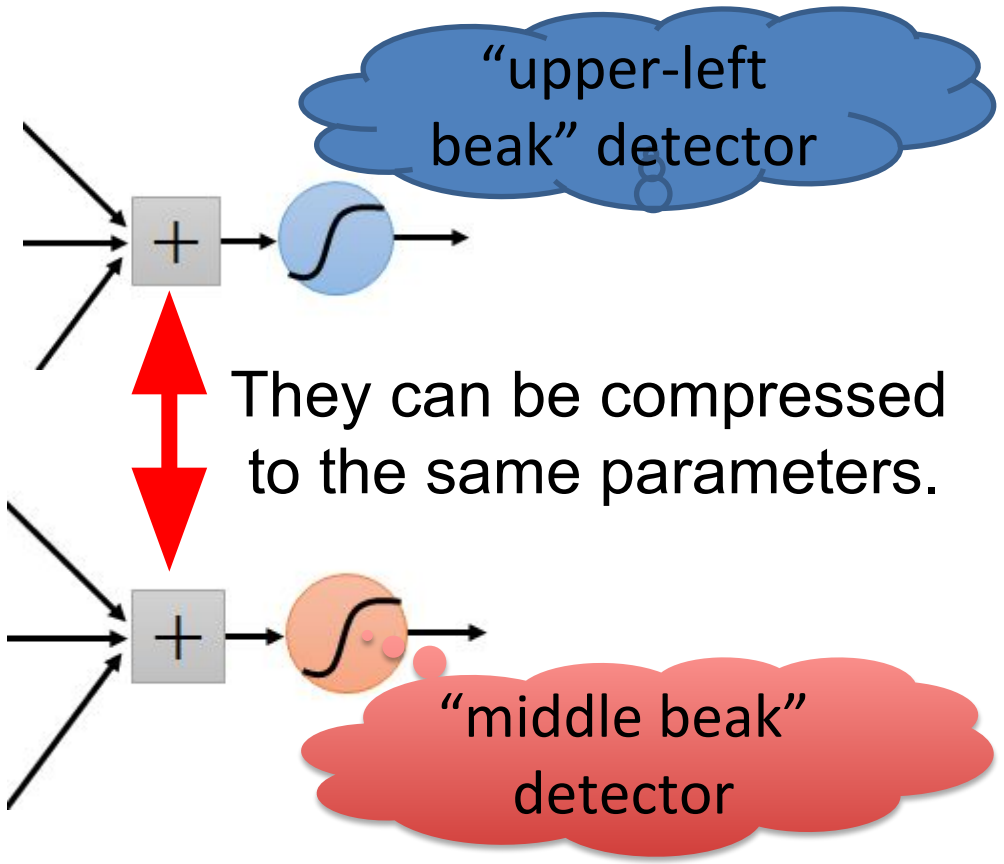
Can represent a small region with fewer parameters



Same pattern appears in different places:
They can be compressed!
What about training a lot of such “small” detectors and each detector must “move around”.

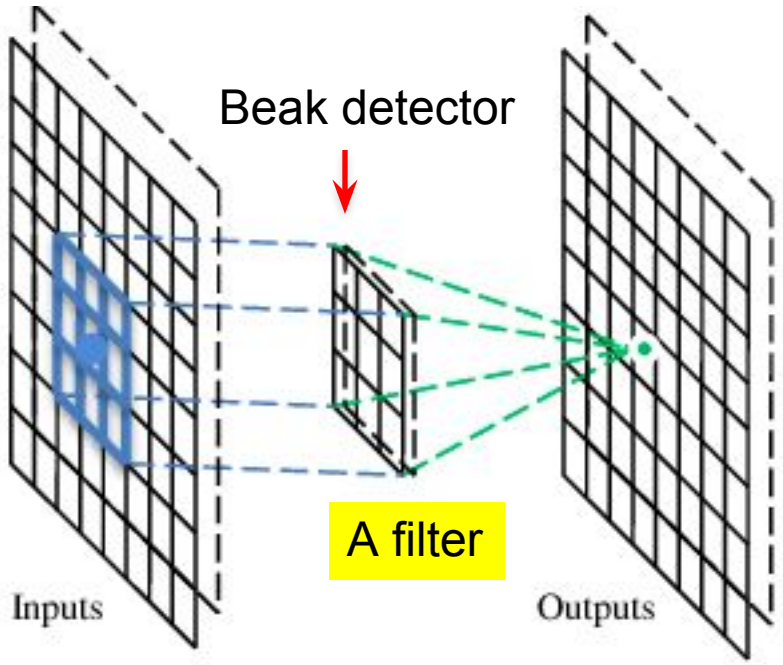






A convolutional layer

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that does convolutional operation.



Convolution

These are the network parameters to be learned.

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

stride= 1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Dot product



6 x 6 image

Convolution

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

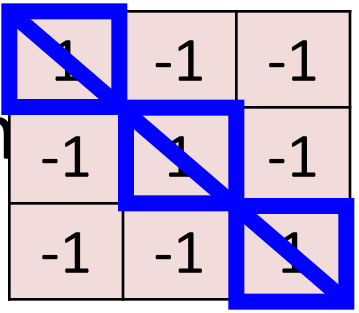
If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

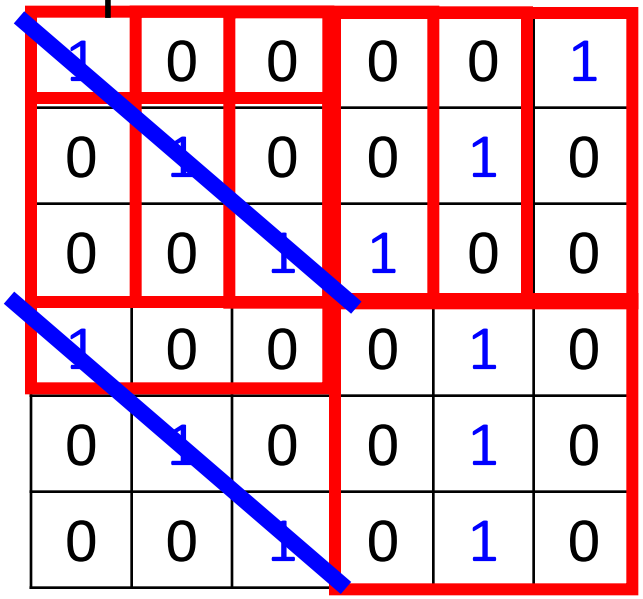
3 -3

Convolution

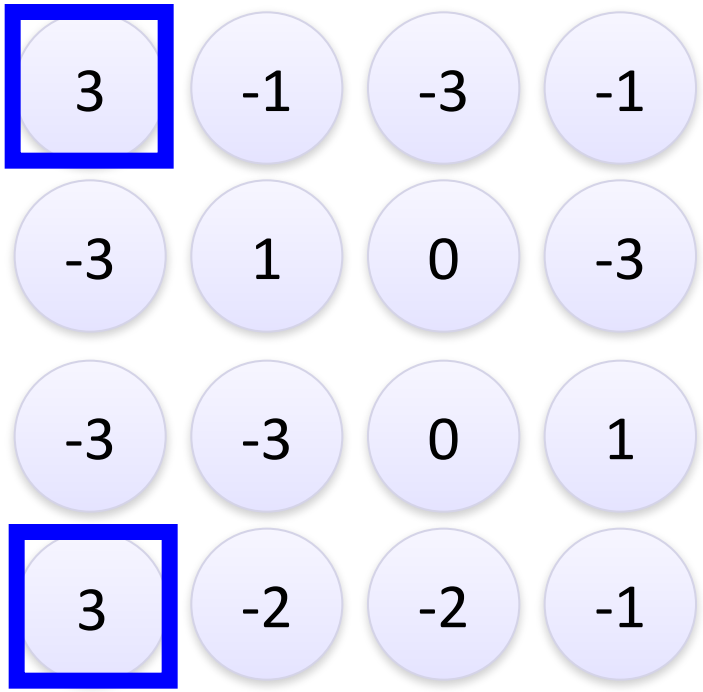


Filter 1

stride= 1



6 x 6 image



Convolution

-1	1	-1
-1	1	-1
-1	1	-1

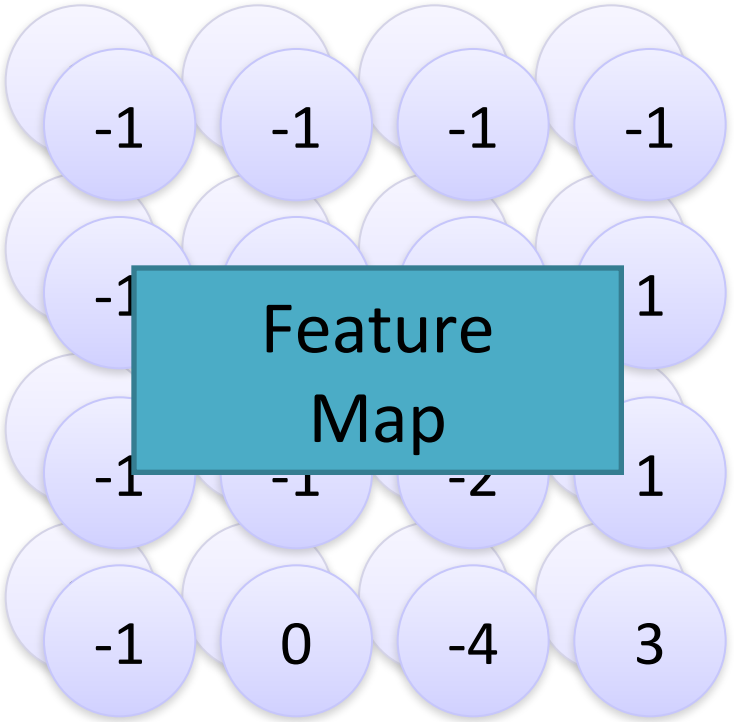
Filter 2

stride= 1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

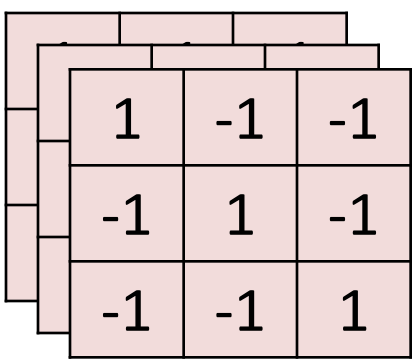
6 x 6 image

Repeat this for each filter

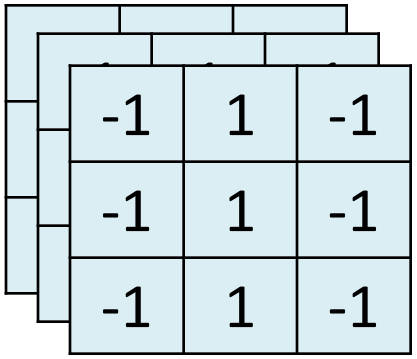


Two 4 x 4 images
Forming 2 x 4 x 4 matrix

Color image: RGB 3 channels

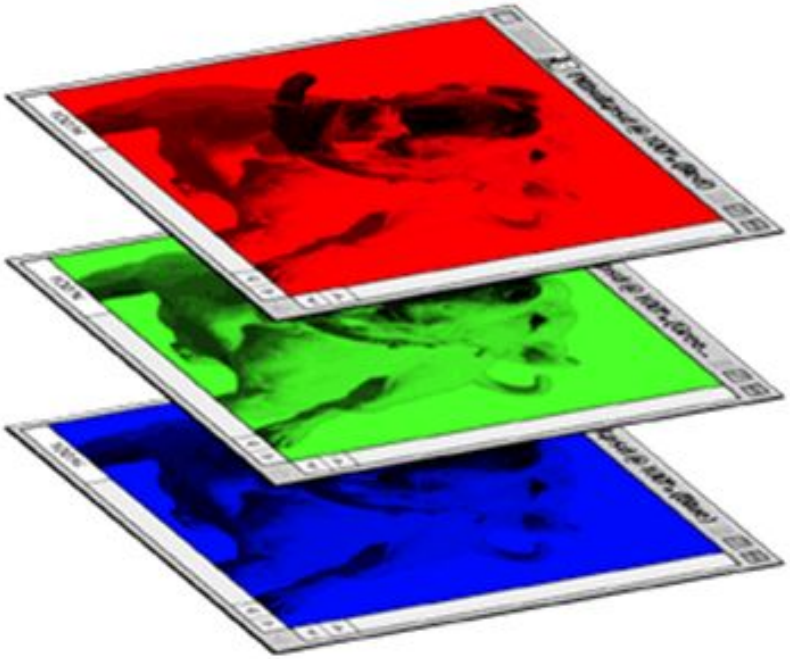


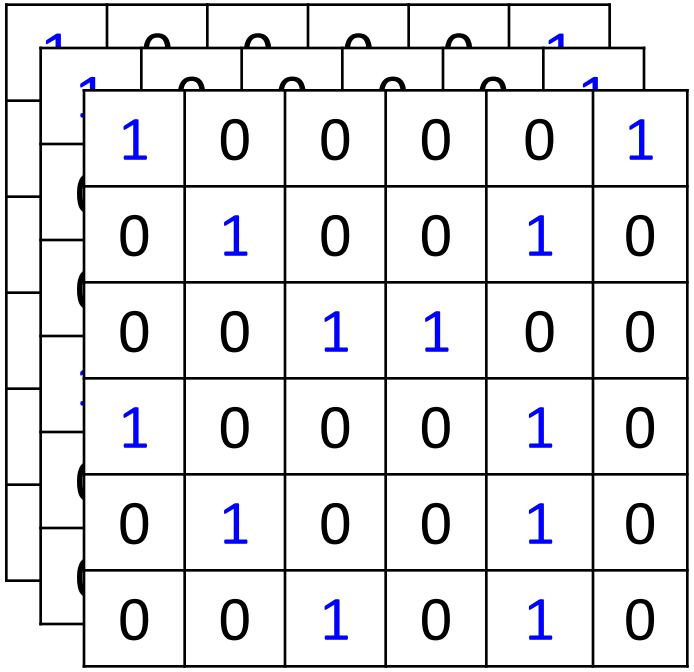
Filter 1



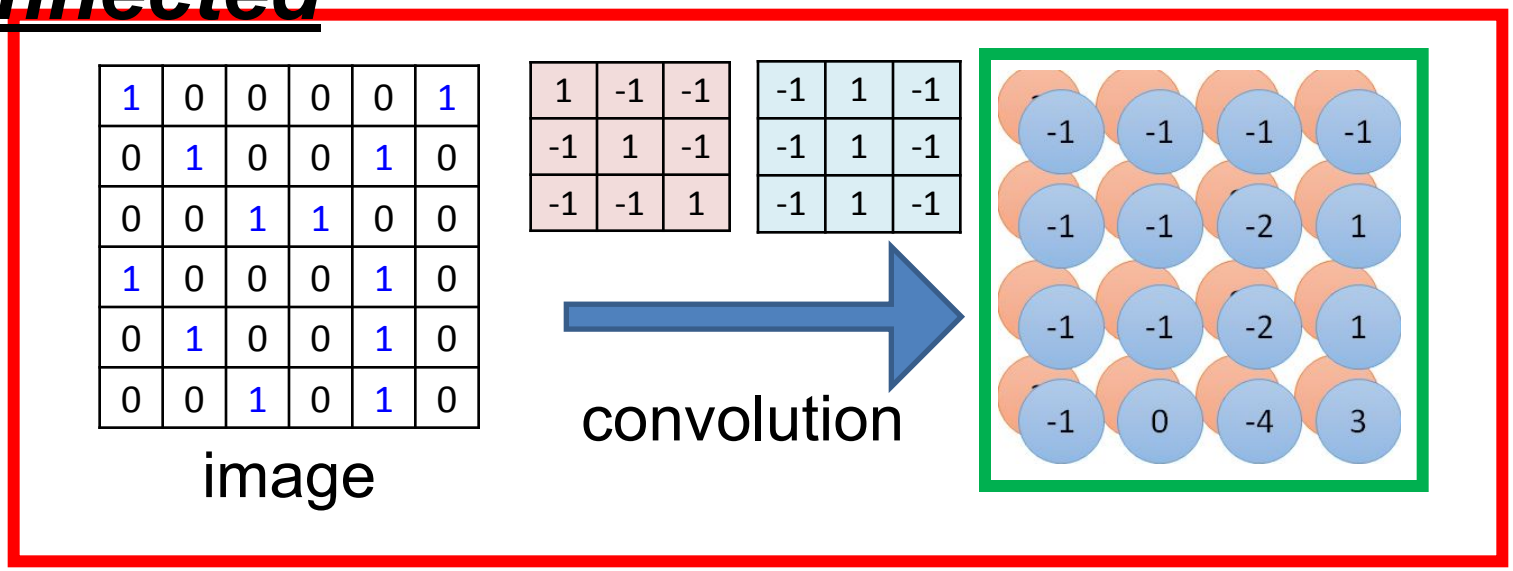
Filter 2

Color image

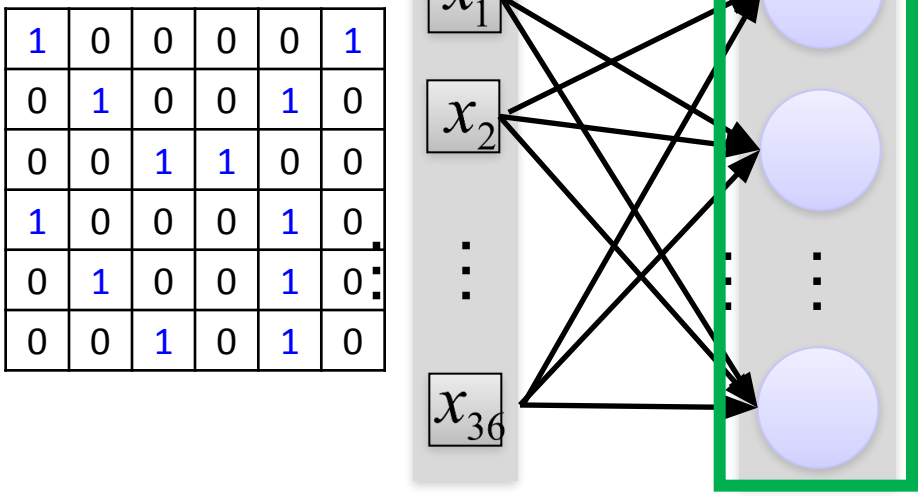


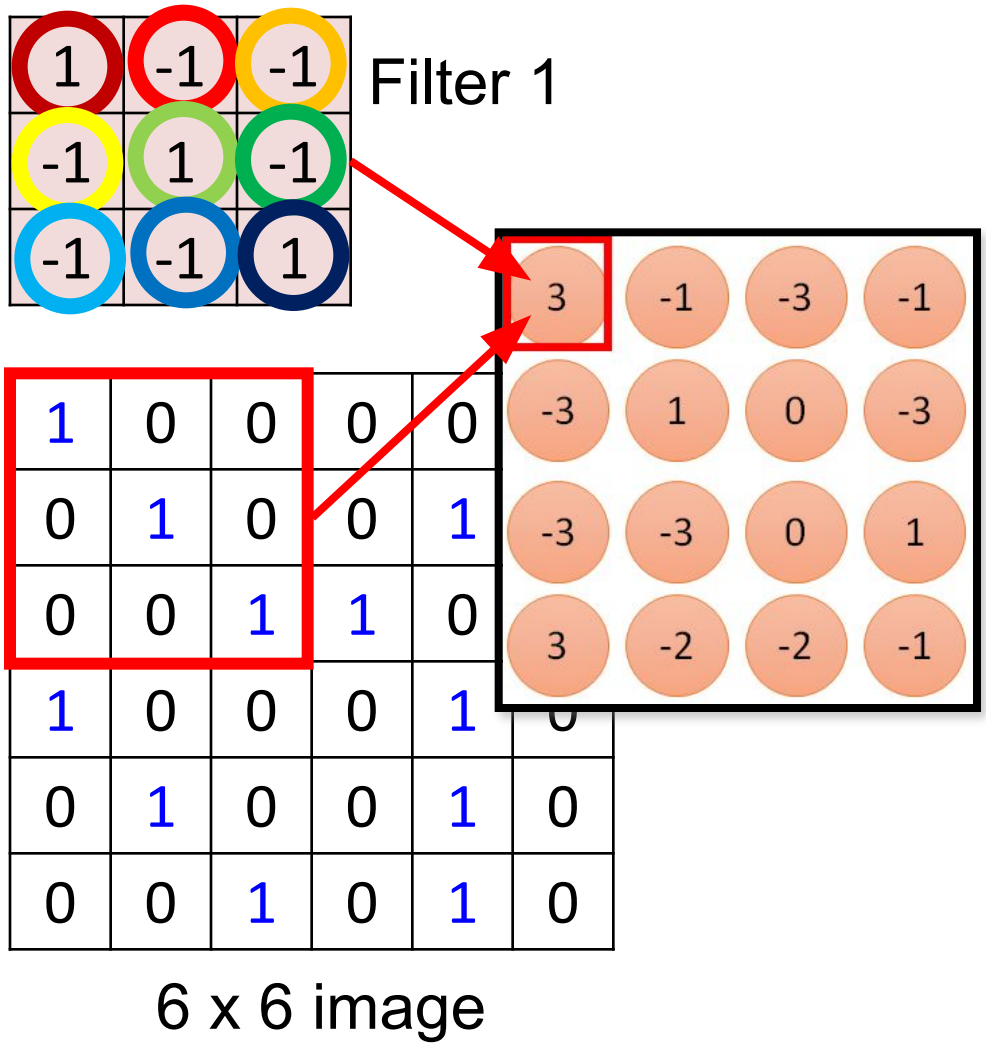


Convolution v.s. Fully Connected

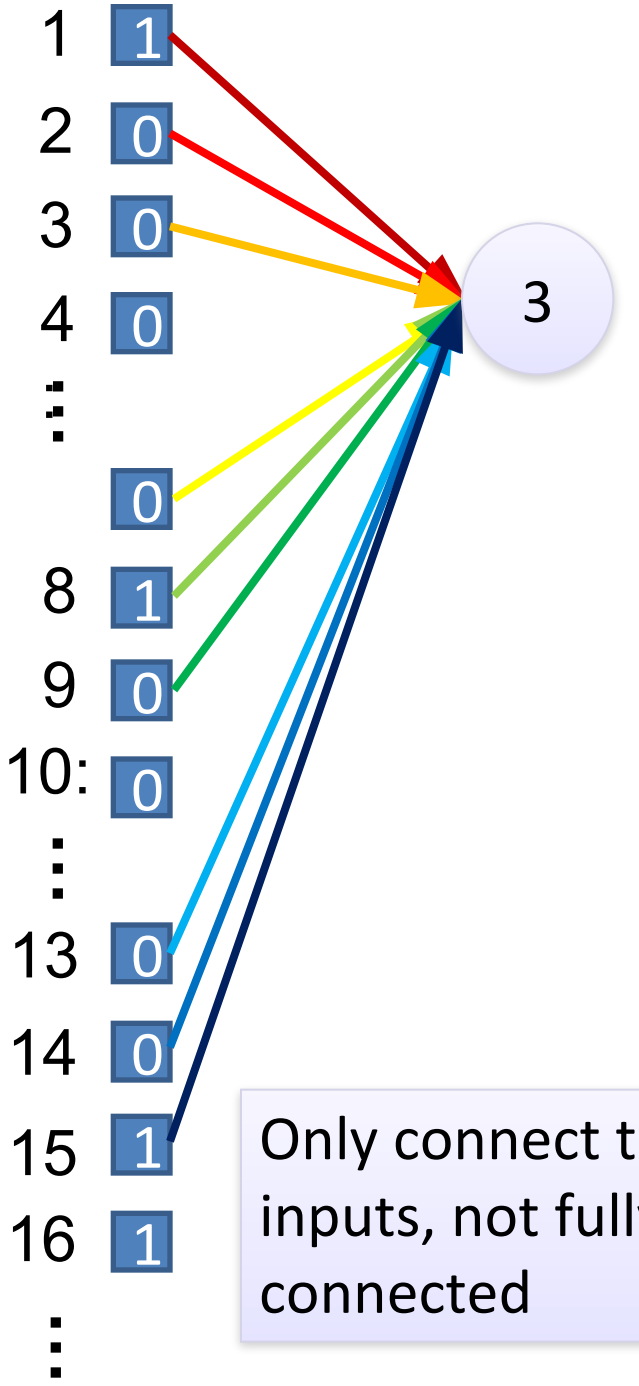


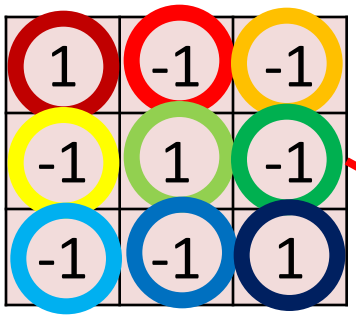
Fully-connected



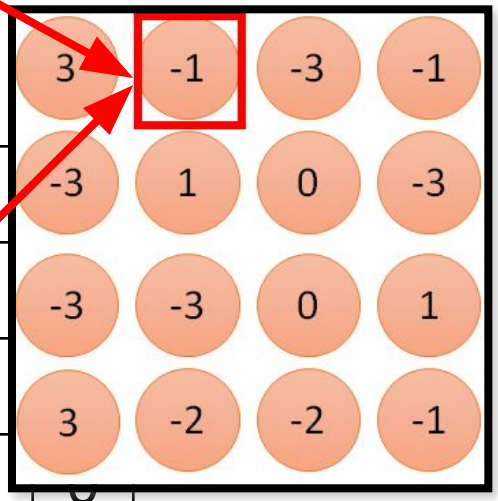


fewer parameters!





Filter 1

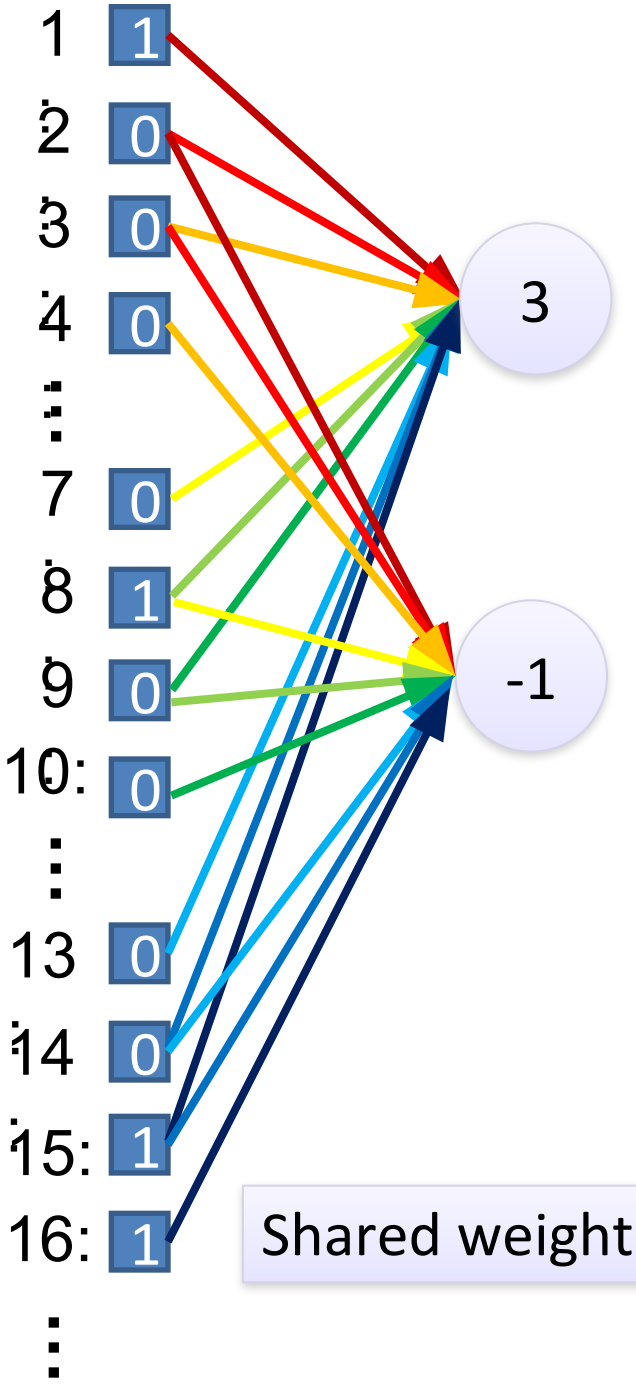


1	0	0	0	0	
0	1	0	0	1	
0	0	1	1	0	
1	0	0	0	1	
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

Fewer parameters

Even fewer parameters

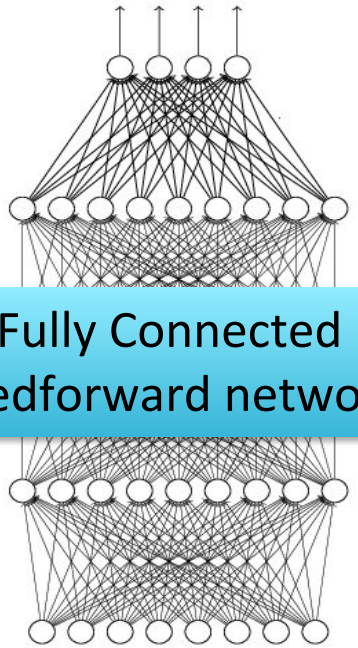


Shared weights

The whole C



cat dog ……



Fully Connected Feedforward network

Convolution

Max Pooling

Convolution

Max Pooling

Can repeat many times

Flattened

Max Pooling

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

3	-1
-3	1

-3	-1
0	-3

-3	-3
3	-2

0	1
-2	-1

-1	-1
-1	-1

-1	-1
-2	1

-1	-1
-1	0

-2	1
-4	3

Why Pooling

- Subsampling pixels will not change the object

bird



bird



Subsampling

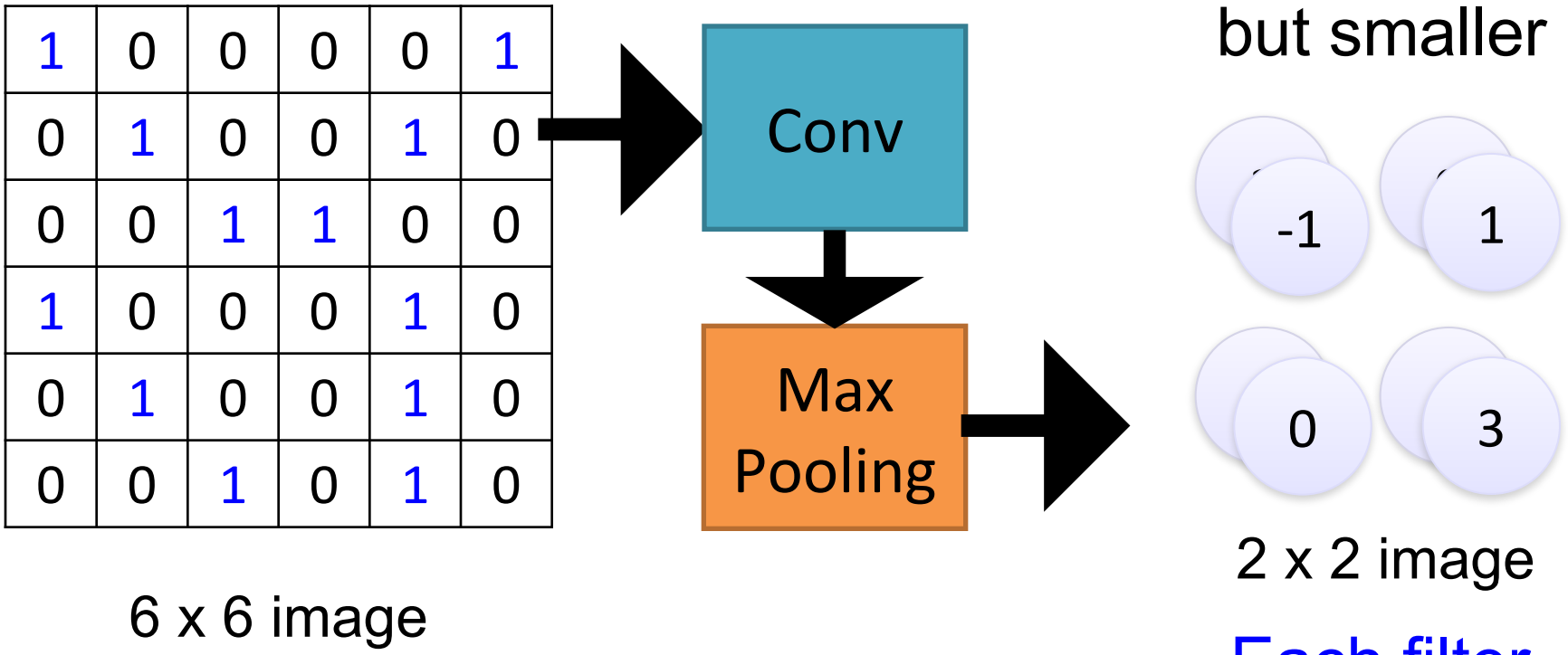
We can subsample the pixels to make image smaller

fewer parameters to characterize the image

A CNN compresses a fully connected network in two ways:

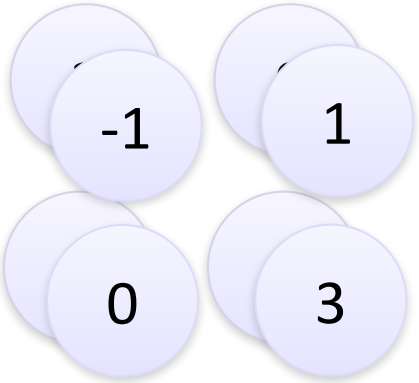
- Reducing number of connections
- Shared weights on the edges
- Max pooling further reduces the complexity

Max Pooling


1 0 0 0 0 1
0 1 0 0 1 0
0 0 1 1 0 0
1 0 0 0 1 0
0 1 0 0 1 0
0 0 1 0 1 0
Conv
Max Pooling
New image but smaller
-1
1
0
3
6 x 6 image
2 x 2 image
Each filter is a channel

The whole C





A new image

Convolution

Max Pooling

Convolution

Max Pooling

Can repeat many times

Smaller than the original image

The number of channels is the number of filters

The whole C



cat dog ……

Convolution

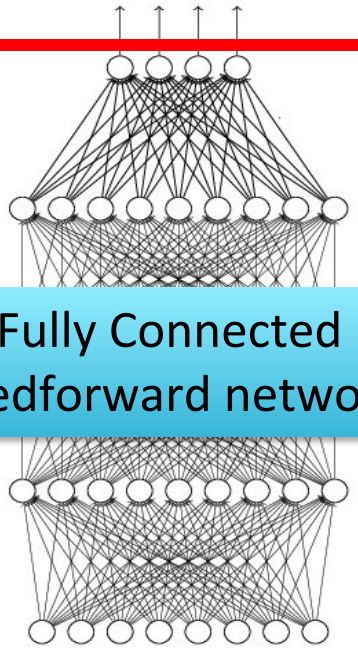
Max Pooling

A new image

Convolution

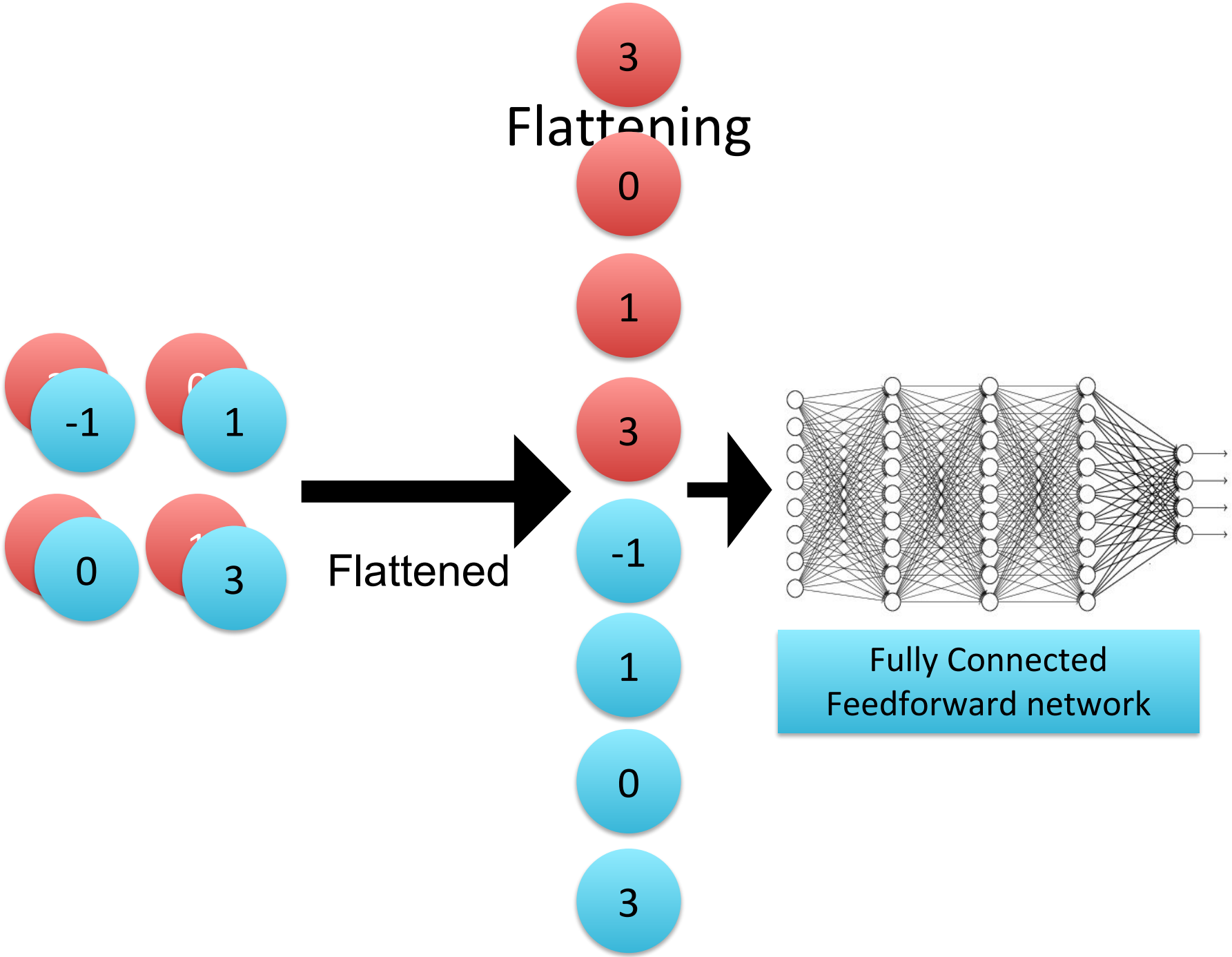
Max Pooling

A new image



Fully Connected Feedforward network

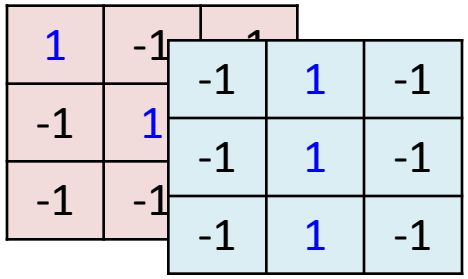
Flattened


Flattening
-1
1
0
3
Flattened
3
0
1
3
-1
1
0
3
Fully Connected
Feedforward network

CNN in Keras

Only modified the ***network structure*** and ***input format (vector -> 3-D tensor)***

```
model2.add( Convolution2D( 25,3,3,
                input_shape=(28,28,1)) )
```



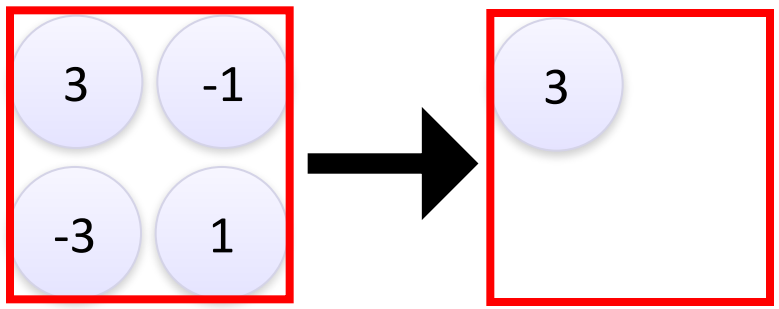
There are 25 3x3 filters.

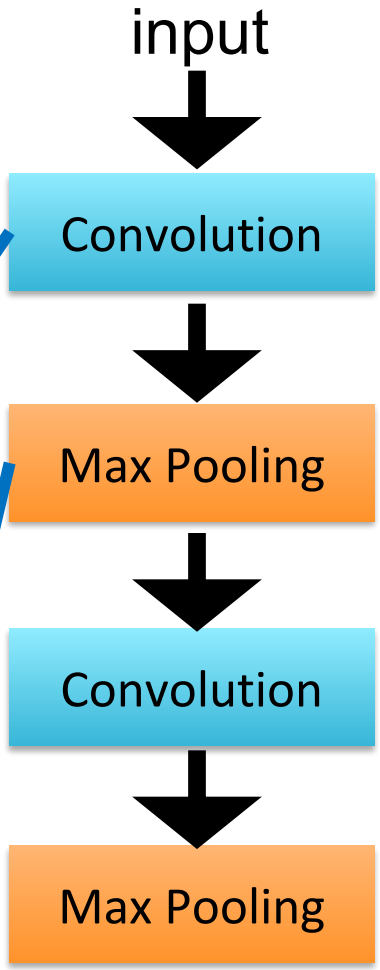
Input_shape = (28 , 28 , 1)

28 x 28 pixels

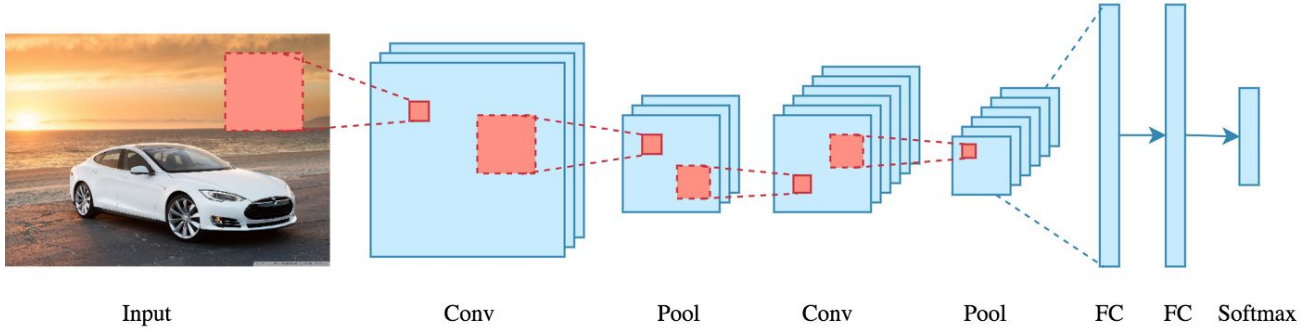
1: black/white, 3: RGB

```
model2.add(MaxPooling2D((2,2)))
```





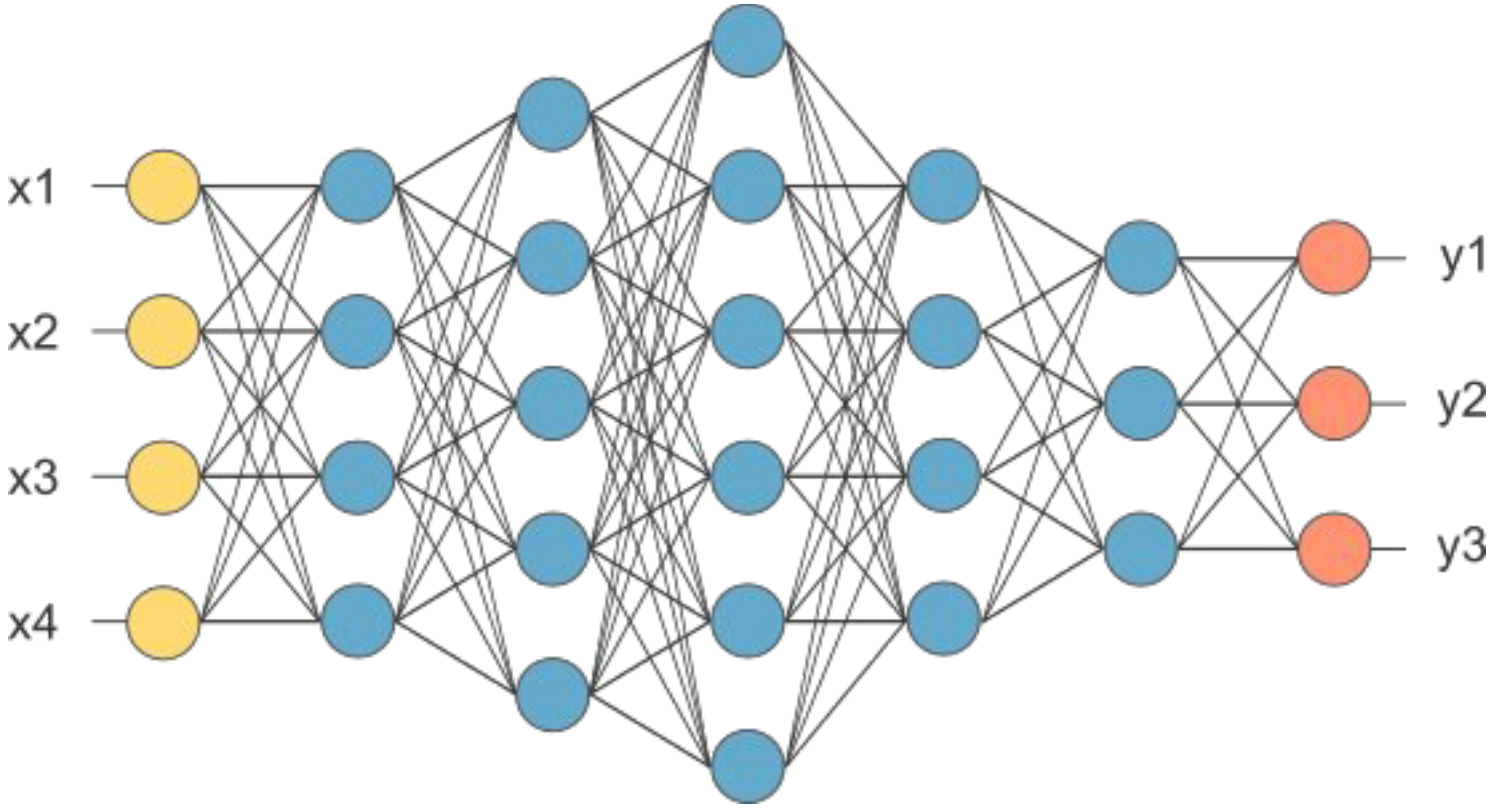
CNN - Architecture



Advantages of CNN…

- Very High accuracy in image recognition problems.
- Automatically detects the important features without any human supervision.
- Weight sharing.

RNN



This is our fully connected network. If x_1 ... x_n, n is very large and growing, this network would become too large. We now will input one x_i at a time, and re-use the same edge weights.

RNN

- A Recurrent Neural Network is a type of neural network that contains loops, allowing information to be stored within the network. In short, Recurrent Neural Networks use their reasoning from previous experiences to inform the upcoming events.

RNN…

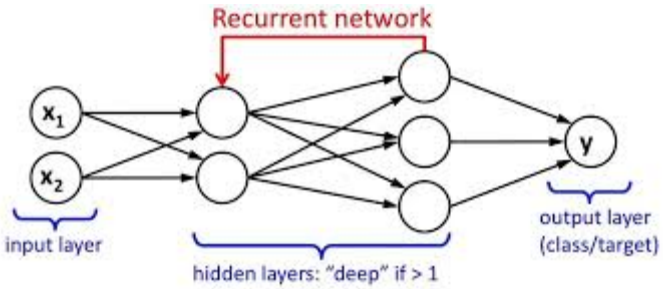
- Recurrent Neural Network (RNN) is a Deep learning algorithm and it is a type of Artificial Neural Network architecture that is specialized for processing sequential data. RNNs are mostly used in the field of Natural Language Processing (NLP).

RNN…..

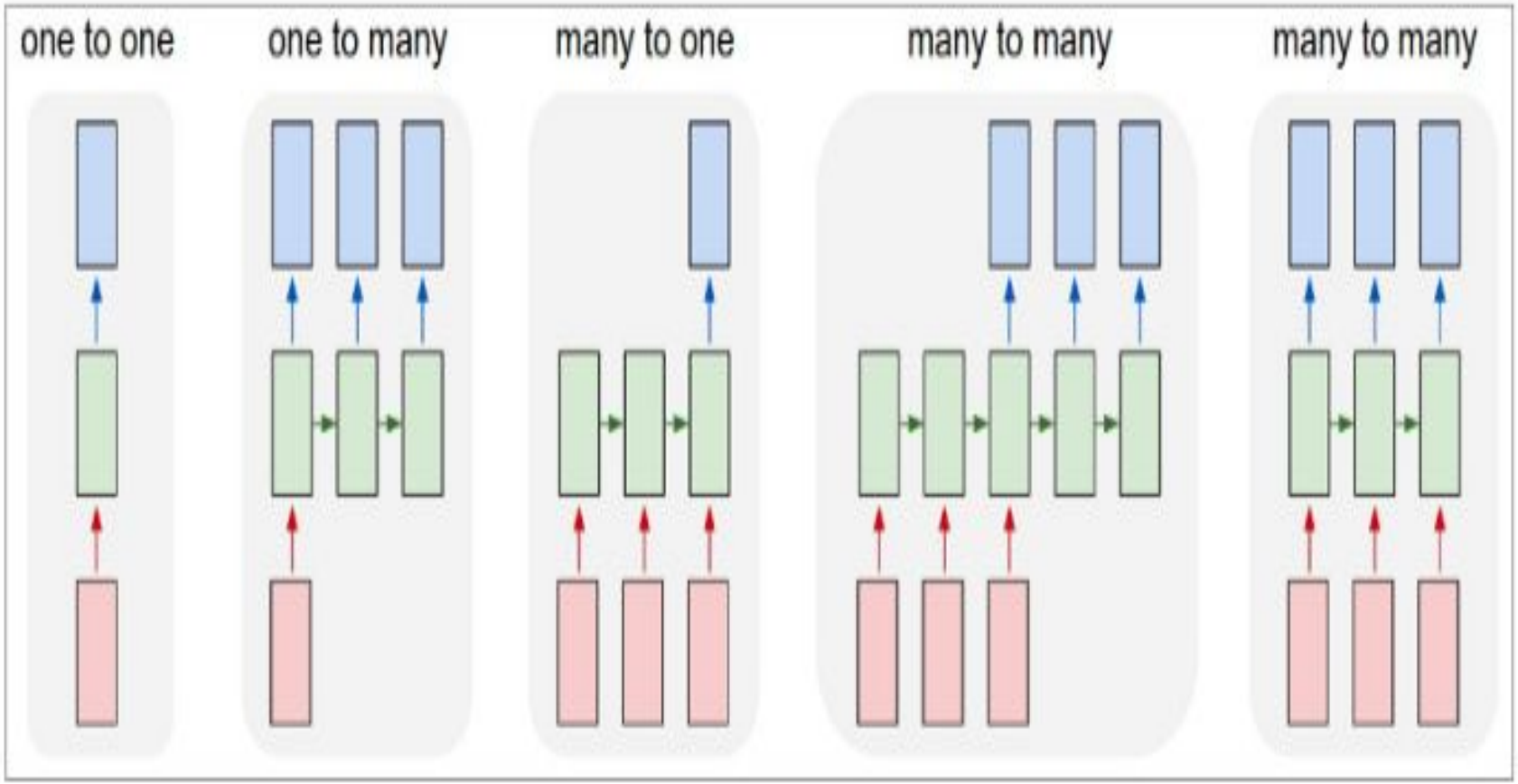
- Recurrent Neural Networks (RNNs) is a popular algorithm used in sequence models.

- Used for **speech recognition, voice recognition, time series prediction, and natural language processing**.

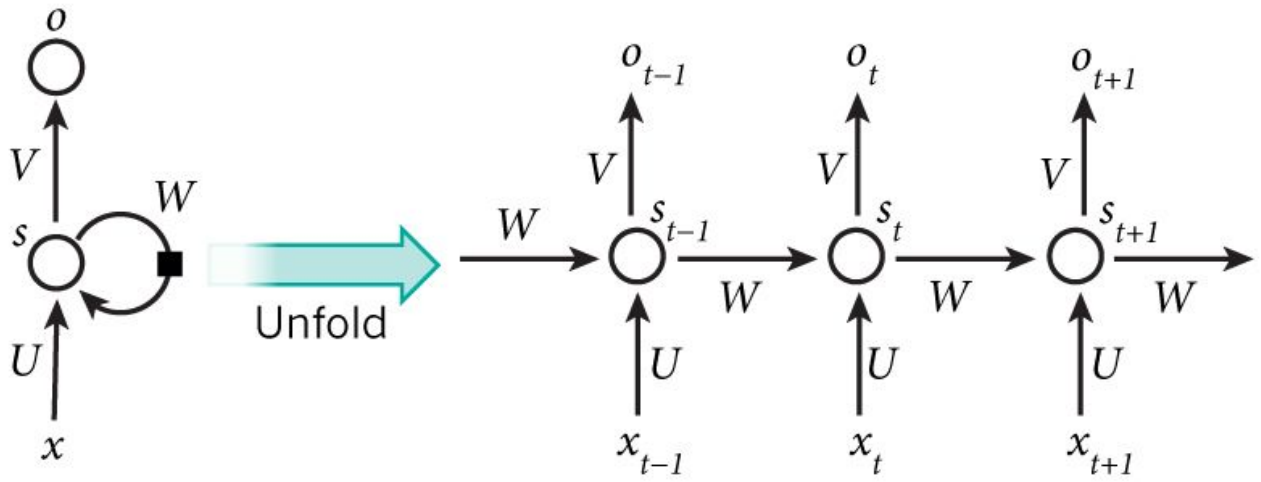
Simple RNN

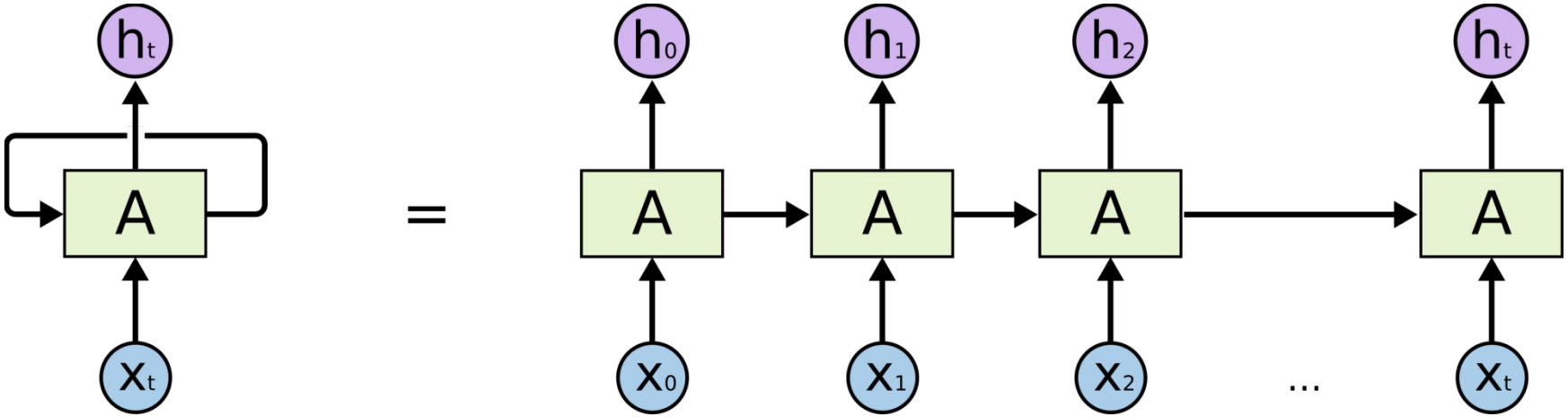

Recurrent network
x1
x2
y
input layer
hidden layers: "deep" if > 1
output layer
(class/target)

How do Recurrent Neural Networks work



Recurrent Neural Network

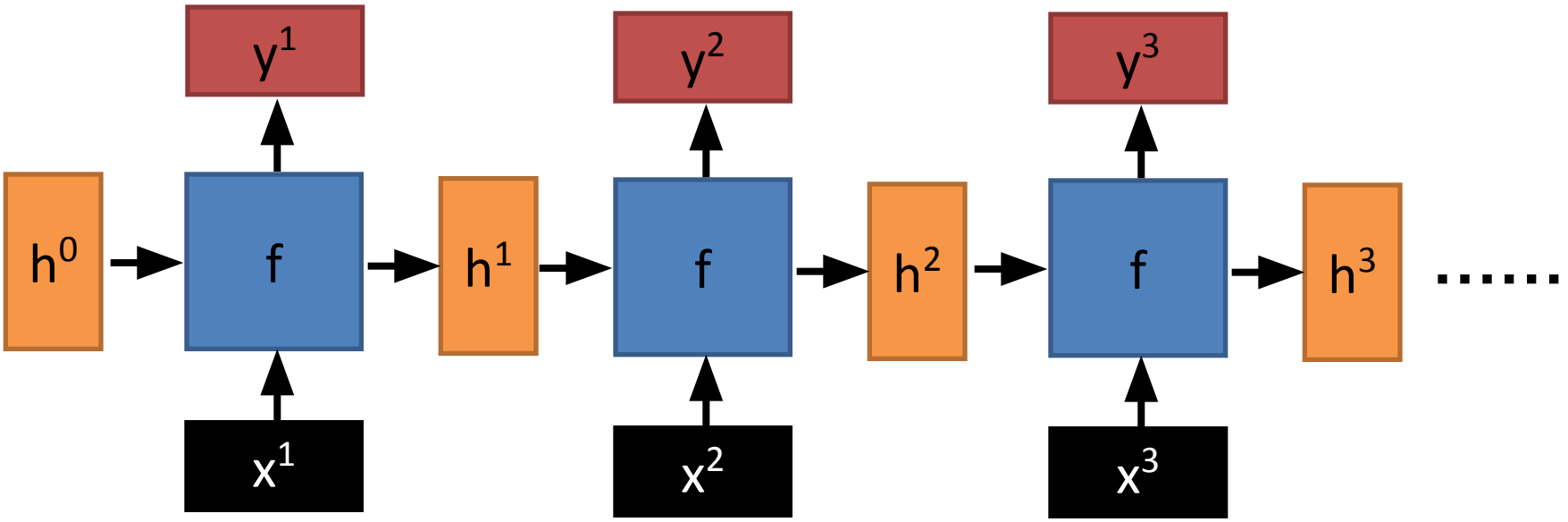




How does RNN reduce complexity?

- Given function f: h',y=f(h,x)

h and h' are vectors with the same dimension



No matter how long the input/output sequence is, we only need one function f. If f's are different, then it becomes a feedforward NN. This may be treated as another compression from fully connected network.

Long Term Short Term Memory (LSTM)

- An RNN has short term memory. When used in combination with Long Short Term Memory (LSTM) Gates, the network can have long term memory.
- Instead of the recurring section of an RNN, an LTSM is a small neural network consisting of four neural network layers. These are the recurring layer from the RNN with three networks acting as gates.

LSTM…….

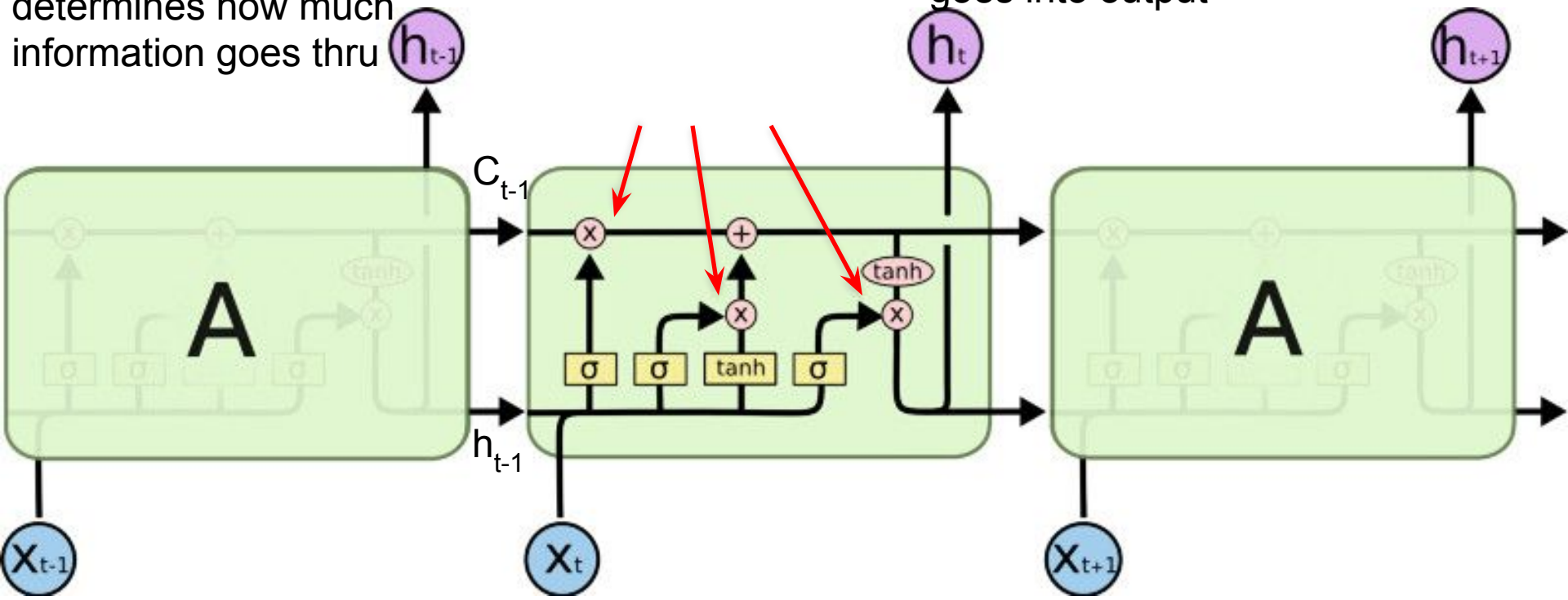
- Long Short Term Memory (LSTM) has three gates.
- An Input gate, this controls the information input at each time step.
- An Output gate, this controls how much information is outputted to the next cell or upward layer
- A Forget gate, this controls how much data to lose at each time step.

This decides what info
Is to add to the cell state

Output gate
Controls what
goes into output

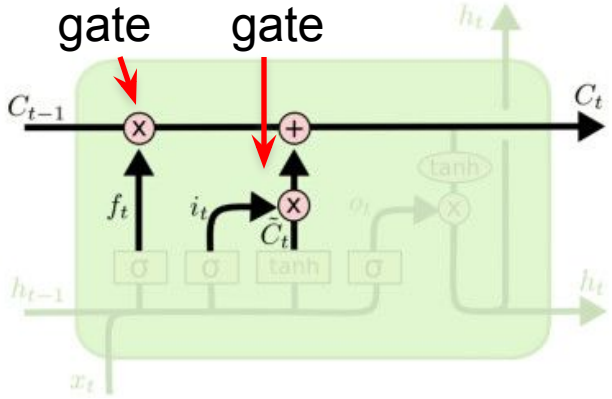
LSTM

This sigmoid gate determines how much information goes thru



Forget gate

input gate



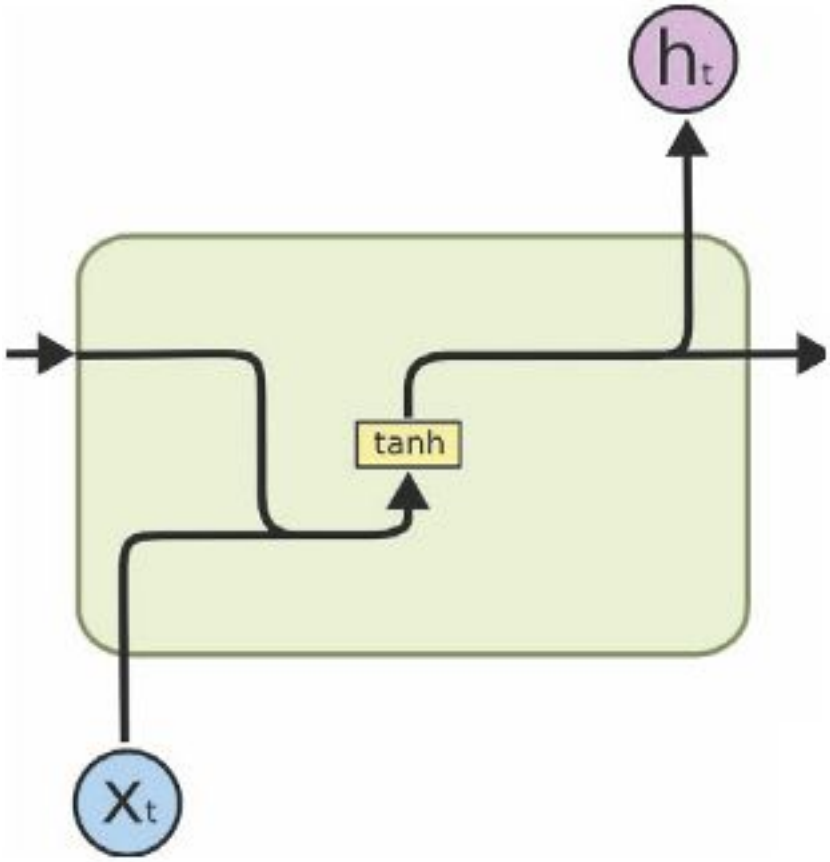
The core idea is this cell state C_t, it is changed slowly, with only minor linear interactions. It is very easy for information to flow along it unchanged.

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

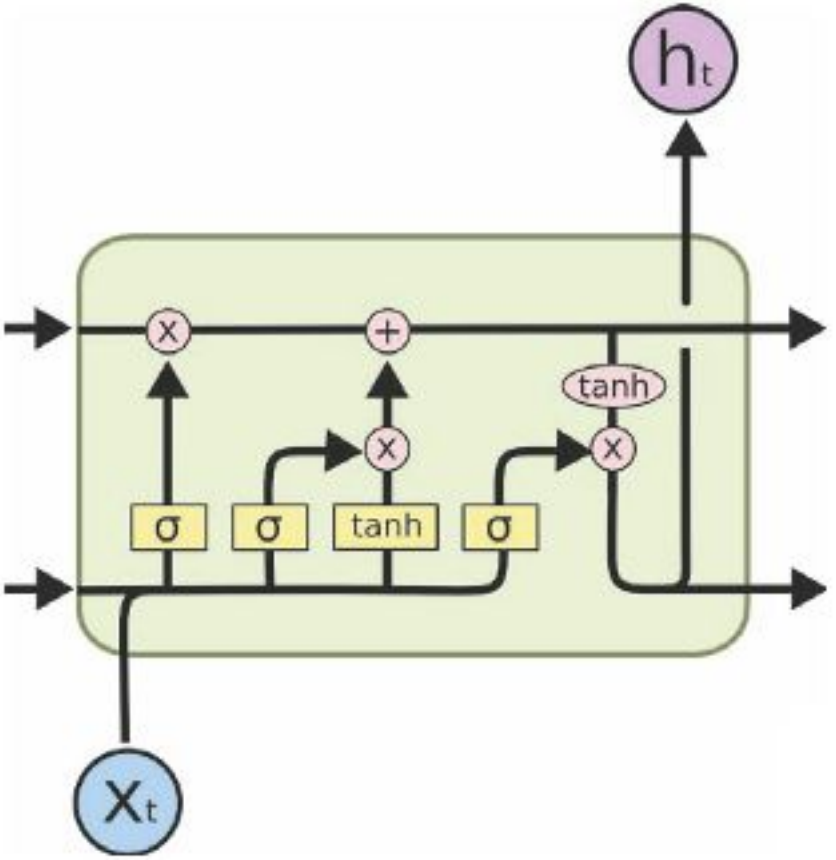
Working of LSTM

- Hidden state and new Input
- Forget gate
- Input gate
- Update cell state
- Update hidden state

RNN vs LSTM



(a) RNN



(b) LSTM

LSTM vs RNN

Both process sequences step-by-step, but LSTM has controlled memory.

Feature	RNN	LSTM
Memory	Simple hidden state	Cell state + hidden state
Long dependencies	Difficult	Better handled
Gates	No	Forget, Input, Output
Complexity	Lower	Higher
Typical use	Shorter patterns	Longer sequence context

Where is LSTM used?

Sequence problems where previous information can influence the current output.

1	**Text / NLP**	Language modeling, sentiment, sequence labeling
2	**Speech**	Speech recognition and audio sequences
3	**Time Series**	Demand, sensor and financial sequences
4	**Healthcare**	Patient / physiological time-series data
5	**IoT**	Sensor streams and anomaly patterns

LSTM — key takeaways

1. LSTM is an RNN variant
2. Cell state carries long-term information
3. Three gates control information flow
4. Forget → Input → Update → Output
5. Useful for sequential / time-dependent data

LSTM = controlled memory for sequence data

LSTM — key takeaways

1. LSTM is an RNN variant
2. Cell state carries long-term information
3. Three gates control information flow
4. Forget → Input → Update → Output
5. Useful for sequential / time-dependent data

LSTM = controlled memory for sequence data